

Spectral Theorem for Self Adjoint and Unitary Operators

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Abstract

This write-up explores the fundamental properties of unitary and self-adjoint operators within the framework of Functional Analysis. We begin with an introduction to these operators and then prove the Spectral Theorem for such operators. We assume the reader is familiar with concepts of operators, adjoints, and functional calculus as found in a first course of Functional Analysis.

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1. Preliminaries about Self Adjoint and Unitary Operators

Definition 1.1. An operator $A \in \mathcal{L}(\mathcal{H})$ where $\mathcal{H}$ is Hilbert space is said to be self adjoint if $A = A^*$. Here, A^* represents the adjoint of A .

Example 1.2. The operator $T \in \mathcal{L}(\mathcal{V})$ defined by $T = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}$ is a self adjoint operator.

Definition 1.3. An operator $U \in \mathcal{L}(\mathcal{H})$ where $\mathcal{H}$ is Hilbert space is said to be unitary if $UU^* = I$.

Example 1.4. The operator $T \in \mathcal{L}(\mathcal{V})$ defined by $T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is a unitary operator.

Proposition 1.5. If $A \in \mathcal{L}(\mathcal{H})$ is self adjoint and bounded, then $\sigma(A) \subset \mathbf{R}$ and in particular,

$$\sigma(A) \subset [-\|A\|, \|A\|]$$

Proof:

Let $\lambda, \mu \in \mathbf{R}$. Then for $u \in \mathcal{H}$,

$$\|(A - (\lambda + i\mu))u\|^2 = \|(A - \lambda)u\|^2 + \mu^2\|u\|^2 \quad (1)$$

Thus, for any $u \in \mathcal{H}$,

$$\|(A - (\lambda + i\mu))u\|^2 \geq \mu^2\|u\|^2 \quad (2)$$

Thus, if $\mu \neq 0$, then $A - (\lambda + i\mu)$ is injective.

Suppose, by contradiction that it is not bijective. Then, $\text{Im}(A - (\lambda + i\mu)) \neq \mathcal{H}$. Then,

$$\text{Ker}(A^* - (\lambda - i\mu)) = \text{Ker}((A - (\lambda + i\mu))^*) = (\text{Im}(A - (\lambda + i\mu)))^\perp \neq 0 \quad (3)$$

which means that it is non invertible and so

$$\lambda - i\mu \in \sigma(A^*)$$

However, A is self adjoint which mean $\sigma_p(A) = \sigma_p(A^*)$

Also, by (2) and $\lambda - i\mu \notin \sigma_p(A)$ for $\mu \neq 0$, we arrive at a contradiction.

Thus, $A - (\lambda + i\mu)$ is bijective whenever $\mu \neq 0$ and if $\mu = 0$, then $\lambda + i\mu \in \rho(A)$ where $\rho(A)$ denotes the resolvent set of A .

Finally, since $|\lambda| \leq \|A\|$,

$$\sigma(A) \subset [-\|A\|, \|A\|] \quad (4)$$

Proposition 1.6. If $U \in \mathcal{L}(\mathcal{H})$ is unitary, then

$$\sigma(U) \subset S^1 = \{z \in \mathbf{C} : |z| = 1\}$$

Proof:

If U is unitary,

$$\|U\| = 1$$

Thus if $|z| > 1$, $z \in \rho(U)$ and so $z - U$ is invertible.

If $|z| > 1$,

$$z - U = -IU + zU^{-1}U = U(I - zU^{-1})$$

and

$$\|zU^{-1}\| = |z|\|U^*\| = |z| < 1$$

So we can apply the Neumann Series Lemma to conclude that $z - U$ is invertible.

2. Functional Calculus

Using functional calculus, we define for any $T \in \mathcal{L}(\mathcal{H})$ and bounded region Ω with a smooth boundary in $\mathbf{C}$ such that $\sigma(T) \subset \Omega^\circ$, if f is holomorphic on a neighbourhood of Ω ,

$$f(T) = \frac{1}{2\pi i} \int_{\partial\Omega} f(z)(z - T)^{-1} dz$$

This is well defined because the term $R_z = (z - T)^{-1}$ is holomorphic on $\rho(T)$. Note that this definition is a straightforward analog of Cauchy's Integral Formula.

Lemma 2.1. *If $f(z) = 1$, then $f(T) = I$ and if $f(z) = z$, then $f(T) = T$.*

Proof:

Suppose $f(z) = 1$. Then,

$$f(T) = \frac{1}{2\pi i} \int_{\gamma} f(z)(z - T)^{-1} dz = \frac{1}{2\pi i} \int_{\gamma} (z - T)^{-1} dz$$

Thus by a direct analog of Cauchy's Residue Theorem,

$$f(T) = \frac{1}{2\pi i} (2\pi i \cdot \text{Res}(f, T)) = \text{Res}(f, T) = \lim_{z \rightarrow T} (z - T) \frac{1}{z - T} = 1$$

Now suppose $f(z) = z$. Then,

$$\begin{aligned} f(T) &= \frac{1}{2\pi i} \int_{\gamma} z(z - T)^{-1} dz = \frac{1}{2\pi i} \int_{\gamma} z(z^{-1}(I - z^{-1}T)^{-1}) dz \\ &= \frac{1}{2\pi i} \int_{\gamma} (I - z^{-1}T)^{-1} dz = \frac{1}{2\pi i} \int_{\gamma} I + \sum_{n=0}^{\infty} (z^{-1}T)^n dz \\ &= 0 + \frac{1}{2\pi i} \int_{\gamma} \sum_{n=0}^{\infty} \left(\frac{T}{z}\right)^n dz = \frac{1}{2\pi i} \sum_{n=0}^{\infty} \int_{\gamma} \left(\frac{T}{z}\right)^n dz \\ &= \frac{1}{2\pi i} \sum_{n=0}^{\infty} T^n \int_{\gamma} \frac{1}{z^n} dz \\ &= \begin{cases} 2\pi i & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases} \tag{5} \\ &= \frac{1}{2\pi i} T(2\pi i) = T \end{aligned}$$

where (5) follows from the fact that $\frac{1}{z^n}$ has a simple pole at $z = 0$ only when $n = 1$.

Proposition 2.2. *If f and g are holomorphic on a neighbourhood of Ω , then*

$$f(T)g(T) = (fg)(T)$$

Proof:

Let $\gamma = \partial\Omega$ and let γ_1 the boundary of slightly larger region on which f and g are both holomorphic. Then,

$$f(T)g(T) = f(T) \frac{1}{2\pi i} \int_{\gamma_1} g(z)(z - T)^{-1} dz$$

Now, since f is linearly acting on T ,

$$f(T)g(T) = \frac{1}{2\pi i} \int_{\gamma_1} g(z)f(T)(z - T)^{-1} dz \tag{6}$$

$$= \frac{1}{2\pi i} \int_{\gamma_1} g(z) \left(\frac{1}{2\pi i} \int_{\gamma} f(\zeta) (\zeta - T)^{-1} d\zeta \right) (z - T)^{-1} dz \quad (7)$$

$$= \frac{-1}{4\pi^2} \int_{\gamma_1} \int_{\gamma} g(z) f(\zeta) (\zeta - T)^{-1} (z - T)^{-1} d\zeta dz \quad (8)$$

$$= \frac{-1}{4\pi^2} \int_{\gamma_1} \int_{\gamma} (\zeta - z)^{-1} f(\zeta) g(z) (R_z - R_{\zeta}) d\zeta dz \quad (9)$$

By Cauchy's theorem, the term involving R_z as a factor has $d\zeta$ integral as 0. Then for dz ,

$$g(\zeta) = \frac{1}{2\pi i} \int_{\gamma_1} (z - \zeta)^{-1} g(z) dz \quad (10)$$

Thus,

$$f(T)g(T) = \frac{1}{2\pi i} \int_{\gamma} f(\zeta) g(\zeta) R_{\zeta} d\zeta$$

which gives us the desired conclusion.

2.1. Functions of Unitary Operators

Let $U \in \mathcal{L}(\mathcal{H})$ be unitary. Then from Proposition 1.6,

$$\sigma(U) \subset S^1$$

If Ω is an open neighbourhood of S^1 containing $\{z \in \mathbf{C} : 1 - \epsilon \leq |z| \leq 1 + \epsilon\}$, then for f ,

$$f(U) = \frac{1}{2\pi i} \int_{\partial\Omega} f(z) (z - U)^{-1} dz \quad (11)$$

$$= \frac{1}{2\pi i} \left[\int_{|z|=1+\epsilon} f(z) z^{-1} \sum_{k=0}^{\infty} z^{-k} U^k dz - \int_{|z|=1-\epsilon} f(z) (-U) \sum_{l=0}^{\infty} z^l U^{-l} dz \right] \quad (12)$$

where (12) follows from the Neumann Series Lemma for the cases $|z| > 1$ and $|z| < 1$ in the following way: when $|z| = 1 + \epsilon$, then

$$(z - U)^{-1} = (z(I - U z^{-1}))^{-1} = z^{-1} \sum_{k=0}^{\infty} U^k z^{-k}$$

and when $|z| = 1 - \epsilon$, then we can directly apply the lemma to

$$(z - U)^{-1} = -U(I - zU^{-1})^{-1}$$

Now, let $z = (1 \pm \epsilon)e^{i\theta}$. Then,

$$dz = i(1 \pm \epsilon)e^{i\theta} d\theta$$

So,

$$\begin{aligned} 2\pi i f(U) &= \int_0^{2\pi} \frac{f((1 + \epsilon)e^{i\theta})}{(1 + \epsilon)e^{i\theta}} \sum_{k=0}^{\infty} \frac{U^k i(1 + \epsilon)e^{i\theta}}{(1 + \epsilon)^k e^{ik\theta}} d\theta - \int_0^{2\pi} \frac{f((1 - \epsilon)e^{i\theta})(-U)}{(1 - \epsilon)e^{i\theta}} \sum_{l=0}^{\infty} (1 - \epsilon)^l e^{il\theta} i(1 - \epsilon)e^{i\theta} U^{-l} d\theta \\ &= \int_0^{2\pi} f((1 + \epsilon)e^{i\theta}) \sum_{k=0}^{\infty} \frac{U^k i}{(1 + \epsilon)^k e^{ik\theta}} d\theta - \int_0^{2\pi} f((1 - \epsilon)e^{i\theta})(-U) \sum_{l=0}^{\infty} (1 - \epsilon)^l e^{il\theta} i U^{-l} d\theta \quad (13) \end{aligned}$$

Now as we take the limit $\epsilon \rightarrow 0$ in (13), we get

$$2\pi i f(U) = \int_0^{2\pi} f(e^{i\theta}) \sum_{k=0}^{\infty} \frac{U^k i}{e^{ik\theta}} d\theta - \int_0^{2\pi} f(e^{i\theta}) \sum_{l=0}^{\infty} -U e^{il\theta} i U^{-l} d\theta$$

Thus,

$$f(U) = \frac{1}{2\pi} \left[\int_0^{2\pi} f(e^{i\theta}) \sum_{k=0}^{\infty} \frac{U^k}{e^{ik\theta}} d\theta + \int_0^{2\pi} f(e^{i\theta}) \sum_{l=0}^{\infty} e^{il\theta} U^{l-1} d\theta \right]$$

This is clearly in the form of the Fourier expansion along the circle and so we have

$$f(e^{i\theta}) \sim \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{ik\theta}$$

where $\hat{f} = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-ik\theta} d\theta$.

Therefore, we have that

$$f(U) = \sum_{k=-\infty}^{\infty} \hat{f}(k) U^k \quad (14)$$

Thus $f \mapsto f(U)$ extends uniquely from the space $\mathcal{O}$ of holomorphic functions to a larger space, namely $C^\infty(S^1)$.

By continuity, we also have that for $f, g \in C^\infty(S^1)$,

$$(fg)(U) = f(U)g(U)$$

Now, we know that if $g = \bar{f}$, then

$$\hat{g}(k) = \overline{\hat{f}(-k)}$$

Thus,

$$f(U)^* = \left(\sum_{k=-\infty}^{\infty} \hat{f}(k) U^k \right)^* = \sum_{k=-\infty}^{\infty} \overline{\hat{f}(k)} (U^k)^* = \sum_{k=-\infty}^{\infty} \overline{\hat{f}(k)} (U^{-k}) = \sum_{k=-\infty}^{\infty} \hat{f}(-k) (U^{-k}) = \bar{f}(U)$$

In particular, if $f : S^1 \rightarrow \mathbf{R}$ is real valued, then $f(U)$ is self adjoint.

Proposition 2.3. *If $f \in C^\infty(S^1)$ and $f \geq 0$ on S^1 , then $f(U) \geq 0$.*

Proof:

For $\epsilon > 0$, let $f_\epsilon(z) = f(z) + \epsilon I$ where $z \in S^1$.

This is also self adjoint as $f(z)$ takes real values for all $z \in \sigma(U)$.

Thus, $g_\epsilon(U)$ defined as

$$g_\epsilon(z) = f_\epsilon(z)^{\frac{1}{2}}$$

belongs to $C^\infty(S^1)$ and is strictly greater than 0. Thus,

$$\begin{aligned} \forall \epsilon > 0, \quad f(U) + \epsilon I &= f_\epsilon(U) = g_\epsilon(U)^2 \\ &\geq 0 \end{aligned}$$

Thus, $f(U) \geq 0$.

Proposition 2.4. *If $f \in C^\infty(S^1)$ and $|f| \leq M$ on S^1 , then $\|f(U)\| \leq M$*

Proof:

Set $g = |f|^2 = f\bar{f} \in C^\infty(S^1)$. Thus, on S^1 ,

$$M^2 - g \geq 0$$

which implies

$$M^2 - f\bar{f} \geq 0$$

$$\begin{aligned}
&\implies M^2 - f(U)\bar{f}(U) \geq 0 \\
&\implies M^2 - f(U)f^*(U) \geq 0
\end{aligned} \tag{15}$$

Thus for $v \in S^1$,

$$\begin{aligned}
\|f(U)v\|^2 &= \langle f(U)v, f(U)v \rangle = \langle f^*(U)f(U)v, v \rangle \\
&\leq \langle M^2v, v \rangle = M^2\|v\|^2 \\
&\implies \|f(U)\| \leq M
\end{aligned}$$

Here, inequality (15) follows from Prop 2.3.

Proposition 2.5. *If $U \in \mathcal{L}(\mathcal{H})$ is unitary, then $f \mapsto f(U)$ has a unique continuous extension from $C^\infty(S^1)$ to $C(S^1)$. Further, the following properties hold true:*

For $f, g \in C(S^1)$,

1. $\|f(U)\| \leq \sup_{S^1} |f|$
2. $f \geq 0 \implies f(U) \geq 0$
3. $f(U)^* = \bar{f}(U)$
4. $(fg)(U) = f(U)g(U)$

Proof:

Let $f \in C(S^1)$.

Since $C^\infty(S^1)$ is dense in $C(S^1)$, there exists a sequence of functions $\{f_n\} \subset C^\infty(S^1)$ such that

$$\|f_n - f\|_\infty \rightarrow 0$$

as $n \rightarrow \infty$.

Define $f(U) = \lim_{n \rightarrow \infty} f_n(U)$.

Now, since $f_n \in C^\infty(S^1)$,

$$\|f_n(U) - f_m(U)\|_{\mathcal{L}(\mathcal{H})} \leq \|f_n - f_m\|_\infty$$

Thus, as $\{f_n\}$ is Cauchy in $C(S^1)$ in the $\|\cdot\|_\infty$ norm, $\{f_n(U)\}$ is Cauchy in $\|\cdot\|_{\mathcal{L}(\mathcal{H})}$ norm. Since $\mathcal{L}(\mathcal{H})$ is complete, $f_n(U)$ converges and in fact by uniqueness of limits,

$$f_n(U) \rightarrow f(U)$$

Now, for any $f \in C^\infty(S^1)$,

$$\|f(U)\| \leq \sup_{z \in S^1} |f(z)|$$

and since $\|f_n - f\|_\infty$ converges to 0 uniformly, thus,

$$\|f_n(U) - f(U)\| \leq \sup_{z \in S^1} |f_n(z) - f(z)| \rightarrow 0$$

Thus,

$$f \mapsto f(U)$$

is a continuous extension.

Uniqueness of the extension follows from uniqueness of the limit.

Now, to prove the different properties, notice that the proposition allows us to prove the properties directly using a limiting argument due to the continuity of the extension. In this write up, we will only prove one of the properties.

Property: If $f \in C(S^1)$ and $f \geq 0$ on S^1 , then $f(U) \geq 0$

Proof:

For $f \in C^\infty(S^1)$,

$$f \geq 0 \implies f(U) \geq 0$$

or

$$\langle v, f(U)v \rangle \geq 0$$

for any $v \in \mathcal{H}$.

Consider $\{f_n\} \subset C^\infty(S^1)$ such that $\|f_n - f\|_\infty \rightarrow 0$ for $f_n \geq 0$ on S^1 .

Then as

$$f \mapsto f(U)$$

is a continuous extension,

$$\|f_n(U) - f(U)\| \rightarrow 0$$

i.e $f_n \rightarrow f$ in the operator norm.

Since $f_n \geq 0$, then $f_n(U) \geq 0$ and so

$$\langle v, f_n(U)v \rangle \geq 0$$

So as $n \rightarrow \infty$,

$$\langle v, f(U)v \rangle \geq 0$$

i.e $f(U) \geq 0$.

3. The Spectral Theorems

In this section, we will prove the spectral theorems in the following way:

Theorem 3.1. *If $U \in \mathcal{L}(\mathcal{H})$ is unitary, there exists a measure space $(\mathcal{X}, \mathcal{F}, \mu)$, a unitary map $\phi : \mathcal{H} \rightarrow \mathcal{L}^2(\mathcal{X}, \mu)$ and $u \in \mathcal{L}^\infty(\mathcal{X}, \mu)$ such that*

$$\phi U \phi^{-1} f(x) = u(x) f(x)$$

$\forall f \in \mathcal{L}^2(\mathcal{X}, \mu)$ and $|u| = 1$ on $\mathcal{X}$.

We will then use the above theorem to prove the following self adjoint version of the theorem using the Cayley Transformation.

Theorem 3.2. *If $A \in \mathcal{L}(\mathcal{H})$ is self adjoint, there exists a measure space $(\mathcal{X}, \mathcal{F}, \mu)$, a unitary map $\phi : \mathcal{H} \rightarrow \mathcal{L}^2(\mathcal{X}, \mu)$ and $a \in \mathcal{L}^\infty(\mathcal{X}, \mu)$ such that*

$$\phi A \phi^{-1} f(x) = a(x) f(x)$$

$\forall f \in \mathcal{L}^2(\mathcal{X}, \mu)$. Here, a is real valued and $\|a\|_{\mathcal{L}^\infty} = \|A\|$ on $\mathcal{X}$.

3.1. Preliminary Definitions and Results

Let X be a locally compact Hausdorff space and let $\mathcal{B}$ be a σ algebra on X .

Definition 3.3. A measure μ is called outer regular on a Borel set E if

$$\mu(E) = \inf\{\mu(U) : E \subset U, U \text{ open}\}$$

Similarly, it is called inner regular if

$$\mu(E) = \sup\{\mu(K) : K \subset E, K \text{ compact}\}$$

It is said to be regular if it is both inner and outer regular.

Definition 3.4. A Radon measure is a Borel measure that is inner regular on all open sets and locally finite.

Definition 3.5. A linear functional $I : C_c(X) \rightarrow \mathbf{R}$ is said to be positive if for $f \geq 0$,

$$I(f) \geq 0$$

where $C_c(X)$ denotes the space of continuous functions with compact support in X .

Proposition 3.6. *If I is a positive linear functional on $C_c(X)$, then for each compact set $K \subset X$,*

$$\exists C_K : |I(f)| \leq C_K \|f\|_\infty \quad \forall f \in C_c(X) : \text{supp}(f) \subset K$$

Now suppose that μ is a Borel measure on X and let $E \subset \mathcal{B}$.

When μ is finite on compact sets,

$$\int f d\mu < \infty \quad \forall f \in C_c(X)$$

and

$$\int f d\mu \geq 0 \quad \text{whenever } f \geq 0$$

Thus,

$$I : f \mapsto \int f d\mu$$

is a positive linear functional

Theorem 3.7. Riesz Markov Kakutani

If I is a positive linear functional on $C_c(X)$, then there exists a unique Radon measure μ on X such that

$$I(f) = \int f d\mu$$

Let $U \in \mathcal{L}(\mathcal{H})$ be unitary and let $v \in \mathcal{H} : v \neq 0$.

Set $\mu(f) = \mu_{U,v}(f) = \langle f(U)v, v \rangle$ for $f \in C(S^1)$.

From property 2 of Proposition 2.5,

$$f \geq 0 \implies \mu(f) \geq 0$$

Thus,

$$\mu(f) \geq 0$$

Thus, Theorem 3.7 gives us a Radon measure μ on S^1 defined by

$$\mu(f) = \langle f(U)v, v \rangle = \int_{S^1} f d\mu$$

Define $W = W_{U,v} : C(S^1) \rightarrow \mathcal{H}$ by

$$W(f) = f(U)v$$

If $g \in C(S^1)$,

$$\begin{aligned} \langle W(f), W(g) \rangle_{\mathcal{H}} &= \langle f(U)v, g(U)v \rangle \\ &= \langle \bar{g}(U)f(U)v, v \rangle \\ &= \int_{S^1} f \bar{g} d\mu \\ &= \langle f, g \rangle_{\mathcal{L}^2(S^1, \mu)} \end{aligned}$$

Since this is true for all $f, g \in C(S^1)$ and $C(S^1)$ is dense in $\mathcal{L}^2(S^1, \mu)$, W is an isometry.

By uniqueness of the continuous extension of isometry W , W can be extended to a bounded operator

$$W : \mathcal{L}^2(S^1, \mu) \rightarrow \mathcal{H}$$

and the range of W is the closure in $\mathcal{H}$ of $\mathcal{C}(U, v) = \text{span}\{U^k v : k \in \mathbf{Z}\}$ where $\mathcal{C}(U, v)$ is the cyclic subspace of U containing $v \in \mathcal{H}$ i.e the smallest closed subspace of $\mathcal{H}$ which is invariant under U and contains v .

Definition 3.8. If $\overline{\mathcal{C}(U, v)} = \mathcal{H}$, then we say that v is a cyclic vector for U .

3.2. Spectral Theorem for Unitary Operators

Proposition 3.9. If $U \in \mathcal{L}(\mathcal{H})$ is unitary and has a cyclic vector v , then

$$W : \mathcal{L}^2(S^1, \mu) \rightarrow \mathcal{H}$$

is unitary and

$$W^{-1}UWf(z) = zf(z) \quad \forall f \in \mathcal{L}^2(S^1, \mu)$$

Proof: W is unitary from the fact that it is surjective and is an isometry.

Now,

$$W(zf) = (zf)(U)v = U(f(U)v) = UW(f)$$

from application of functional calculus. Hence we get the result.

Proposition 3.10. If $\mathcal{H}$ is separable and $A \in \mathcal{L}(\mathcal{H})$ is self adjoint, then there exist $v_j \in \mathcal{H}$ such that $\overline{\mathcal{C}(A, v_j)}$ are mutually orothogonal subspaces of $\mathcal{H}$, with the span being dense in $\mathcal{H}$.

Proof:

Let $\{w_j : j \in \mathbf{N}\}$ be dense in $\mathcal{H}$ and $w_j \neq 0$ for all j .

Let $v_1 = w_1$ and construct $\overline{\mathcal{C}(A, w_1)} = H_1$.

If $H_1 = \mathcal{H}$, we are done. If not, we proceed as follows:

To continue, let $j \geq 2 : w_j \notin H_1$ and let v_2 denote the orthogonal projection of w_j onto $H_1^\perp$.

Set $H_2 = \overline{\mathcal{C}(A, v_2)} \subset H_1^\perp$. Clearly,

$$H_1 \oplus H_2 \supset \text{span}\{w_k : 1 \leq k \leq j\}$$

If $H_1 \oplus H_2 = \mathcal{H}$, we are done. Other wise take $j_3 > j : w_{j_3} \notin H_1 \oplus H_2$ and let v_3 be the orthogonal projection of w_{j_3} onto $(H_1 \oplus H_2)^\perp$ and set $H_3 = \overline{\mathcal{C}(A, v_3)} \subset (H_1 \oplus H_2)^\perp$

We then continue similarly.

If for some K , $H_1 \oplus \dots \oplus H_K = \mathcal{H}$, then we are done.

If not, we get a countable sequence of orthogonal spaces $H_k = \overline{\mathcal{C}(A, v_k)}$ whose span contains w_j for every $j \in \mathbf{N}$ and is therefore dense in $\mathcal{H}$.

We now prove Theorem 3.1.

Proof:

Let $H = \oplus_{j \geq 1} H_j$ where each $H_j = \overline{\mathcal{C}(U, v_j)}$ and

$$W_j = W_{A, v_j} : C(S^1) \rightarrow H_j$$

given by

$$W_j(f) = f(U)v_j$$

Extending these to unitary maps, we get

$$W_j : \mathcal{L}^2(S^1, \mu_j) \rightarrow H_j$$

where μ_j is the Radon measure associated to the positive linear functional:

$$\mu_j(f) = \langle f(U)v_j, v_j \rangle$$

These satisfy

$$W_j^{-1} U W_j f = x f$$

where $f \in \mathcal{L}^2(S^1, \mu_j)$. Thus we define the measure space (X, μ) as

$$(X, \mu) = \bigsqcup_{j \geq 1} (S^1, \mu_j)$$

and so

$$\mathcal{L}^2(X, \mu) = \oplus_{j \geq 1} \mathcal{L}^2(S^1, \mu_j)$$

Thus, we have found

$$W : \mathcal{L}^2(X, \mu) \rightarrow \mathcal{H}$$

satisfying

$$W^{-1} U W f(x) = u(x) f(x)$$

where $u(x) = x$ and $|u| = 1$.

Remark: In the case where $\mathcal{H}$ is not separable, we can use Zorn's lemma to change Proposition 3.10 to say that we can find a maximal family of mutually orthogonal subspaces $\{H_\alpha\}_{\alpha \in \mathcal{A}}$ such that

$$\mathcal{H} = \bigoplus_{\alpha \in \mathcal{A}} H_\alpha$$

and use this decomposition in the proof of the theorem similarly.

3.3. The Cayley Transform

In this section, we will show how to use a concept known as the Cayley Transform to prove Theorem 3.2 directly from Theorem 3.1.

Let $A \in \mathcal{L}(\mathcal{H})$ be self adjoint.

Then from Proposition 1.5, $\sigma(A) \subset \mathbf{R}$.

Thus, $A \pm i$ are invertible and $(A \pm i)^{-1} \in \mathcal{L}(\mathcal{H})$.

Define the Cayley Transform as

$$U := (A + i)(A - i)^{-1}$$

Since

$$(A - i)(A - i)^{-1} = I = (A - i)^{-1}(A - i)$$

Thus, $A - i$ and $(A - i)^{-1}$ commute with A and so it also commutes with $A + i$
So we can also write

$$U = (A - i)^{-1}(A + i)$$

Further,

$$U^* = ((A + i)(A - i)^{-1})^* = ((A - i)^{-1})^*(A + i)^* = ((A - i)^*)^{-1}(A + i)^* = (A + i)^{-1}(A - i) = U^{-1}$$

Thus, U is unitary and

$$A(U - 1) = i(U + 1) \quad (16)$$

Lemma 3.11. *If $A \in \mathcal{L}(\mathcal{H})$ is self adjoint and U is its Cayley transform, then $1 \in \rho(U)$.*

Proof:

$$\begin{aligned} U &= (A + i)(A - i)^{-1} \\ &= (A + i)(A + i)(A + i)^{-1}(A - i)^{-1} \\ &= (A + i)^2(A^2 + 1)^{-1} \\ &= (A^2 + 2Ai - 1)(A - i)^{-1} \\ &= (A^2 - 1)(A^2 + 1)^{-1} - 2iA(A^2 + 1)^{-1} \end{aligned}$$

Thus,

$$U = (A^2 - 1)(A^2 + 1)^{-1} - 2iA(A^2 + 1)^{-1} \quad (17)$$

The commutativity between $A^2 - 1, A^2 + 1$ and A implies that $A^2 - 1$ and A also commute with $(A^2 + 1)^{-1}$.

Thus we get that $(A^2 - 1)(A^2 + 1)^{-1}$ and $A(A^2 + 1)^{-1}$ are self adjoint.

Now, for every $v \in \mathcal{H}$, (17) gives us

$$\begin{aligned} \operatorname{Re} \langle Uv, v \rangle &= \langle (A^2 - 1)(A^2 + 1)^{-1}v, v \rangle \\ &= \langle (A^2 + 1 - 2)(A^2 + 1)^{-1}v, v \rangle \\ &= \langle (I - 2(A^2 + 1)^{-1})v, v \rangle \\ &= \langle Iv, v \rangle - 2 \langle (A^2 + 1)^{-1}v, v \rangle \\ &= \|v\|^2 - 2 \langle (A^2 + 1)^{-1}v, v \rangle \end{aligned}$$

Thus,

$$\begin{aligned} \operatorname{Re}(\langle (I - U)v, v \rangle) &= 2 \langle (A^2 + 1)^{-1}v, v \rangle \\ &= 2 \langle w, (A^2 + 1)w \rangle \\ &= 2\|w\|^2 + 2\|Aw\|^2 \\ &\geq 2\|w\|^2 \end{aligned} \quad (18)$$

where $w = (A^2 + 1)^{-1}v$. Now,

$$\|v\| = \|(A^2 + 1)w\| \leq C\|w\|$$

with $C = \|A^2 + 1\| < \infty$. From (18), we get

$$\operatorname{Re}(\langle (I - U)v, v \rangle) \geq \frac{2}{C^2}\|v\|^2$$

and so by Cauchy's inequality,

$$\|(I - U)v\| \geq \frac{2}{C^2}\|v\|$$

So we get that

$$I - U : \mathcal{H} \rightarrow \mathcal{H}$$

is injective with a closed range.

Now if $w \in \mathcal{H}$ is orthogonal to the range of $I - U$, then

$$\begin{aligned}\forall v \in \mathcal{H}, \quad 0 &= \langle (I - U)v, w \rangle = \langle v, (I - U^{-1})w \rangle \\ &= -\langle v, U^{-1}(I - U)w \rangle \\ &= -\langle Uv, (I - U)w \rangle\end{aligned}$$

which gives

$$0 = -\langle Uv, (I - U)w \rangle \tag{19}$$

Since $U : \mathcal{H} \rightarrow \mathcal{H}$ is invertible, therefore

$$\begin{aligned}\forall \tilde{v} \in \mathcal{H}, \quad (19) &\implies \langle \tilde{v}, (I - U)w \rangle = 0 \\ &\implies (I - U)w = 0 \\ &\implies w = 0\end{aligned}$$

as $I - U$ is injective. Thus, $\text{Ran}(I - U)^\perp = \{0\}$ and $\text{Ran}(I - U)$ is dense in $\mathcal{H}$. Thus $I - U$ is invertible and $1 \in \rho(U)$.

Now, using the above lemma and (16), we get that

$$A = i(U + 1)(U - 1)^{-1}$$

$(U - 1)^{-1}$ commutes with $U - 1$ and hence with U and with $U + 1$. So,

$$A = i(U - 1)^{-1}(U + 1)$$

as well.

Now,

$$\begin{aligned}A^* &= -i((U - 1)^{-1})^*(U + 1)^* \\ &= -i(U^{-1} - 1)^{-1}(U^{-1} + 1)\end{aligned}$$

and since $(U^{-1} - 1)^{-1} = (1 - U)^{-1}U$, we get

$$A^* = i(U - 1)^{-1}(U + 1) = A$$

and so A is self adjoint.

3.4. Spectral Theorem for Self Adjoint Operators

We use Theorem 3.1 in the following way:

If $U \in \mathcal{L}(\mathcal{H})$ is unitary, there exists a measure space $(\mathcal{X}, \mathcal{F}, \mu)$, a unitary map $\phi : \mathcal{H} \rightarrow \mathcal{L}^2(\mathcal{X}, \mu)$ and $u \in \mathcal{L}^\infty(\mathcal{X}, \mu)$ such that

$$\phi U v(x) = u(x)\phi v(x)$$

$\forall v \in \mathcal{H}$ and $|u| = 1$ on $\mathcal{X}$.

Proof of Theorem 3.2:

From Lemma 3.11, $1 \in \rho(M_u)$ where M_u is the multiplication operator on $\mathcal{L}^2(X, \mu)$.

Thus, $M_u - 1$ is invertible

$$\implies \exists \delta > 0 : |u(x) - 1| \geq \delta$$

for almost every $x \in \mathcal{X}$ since $\sigma(M_u) = \text{Im ess}(u)$.

Thus from the motivation of Cayley's Transform, we define

$$a(x) = i \frac{u(x) + 1}{u(x) - 1}$$

where u is the same as in Theorem 3.1 and so $a \in \mathcal{L}^\infty(\mathcal{X}, \mu)$.

Now,

$$\overline{a(x)} = -i \frac{\overline{u(x)} + 1}{u(x) - 1} = -i \frac{\frac{1}{u(x)} + 1}{\frac{1}{u(x)} - 1}$$

as $|u(x)| = 1$

$$= -i \frac{1 + u(x)}{1 - u(x)} = i \frac{u(x) + 1}{u(x) - 1} = a(x)$$

for almost every $x \in \mathcal{X}$.

Thus we get that

$$\forall v \in \mathcal{H}, \phi Av(x) = a(x)\phi v(x)$$

where

$$Av(x) = a(x)v(x)$$

A is self adjoint because

$$\begin{aligned} \langle Av, w \rangle &= \int_{\mathcal{X}} a(x)v(x)\overline{w(x)}d\mu(x) \\ &= \int_{\mathcal{X}} v(x)\overline{a(x)w(x)}d\mu(x) = \langle v, Aw \rangle \end{aligned}$$

where the second equality follows from the fact that $a(x) = \overline{a(x)}$.

Thus, we have proved Theorem 3.2.

3.5. Applications:

The Spectral Theorem has several applications in other disciplines such as Physics and Engineering. Two important applications are as follows:

Quantum Mechanics: In Quantum Mechanics, the Hamiltonian Operator is an operator that corresponds to the total energy in the system including kinetic and potential energy. This operator is self adjoint so the Spectral Theorem guarantees a real spectrum corresponding to measurable energy levels.

The Spectral Theorem allows the decomposition of quantum states into eigenstates of energy. It helps physicians to interpret probabilities of outcomes in measurements.

The Hamiltonian operator is further used to define the unitary operator

$$U := e^{-iHt/\hbar}$$

which is also known as the Time Evolution operator of a quantum system. This is used in simulations of quantum systems, quantum computing, and atomic physics.

Signal Processing: The Fourier transform maps between different bases of functions while preserving the $\mathcal{L}^2$ inner product, making it a unitary operator. In signal processing, a signal $f(t)$ can be transformed into its frequency components as

$$F(\omega) = \mathcal{F}\{f(t)\} = Uf(t)$$

Here U is unitary as it preserves the energy of the signal.

The Spectral Theorem explains how signals (functions in $\mathcal{L}^2$) decompose into frequency components (eigenfunctions). This is the foundation of applications like noise reduction, image compression, and data analysis.

4. References

- [1] M. Taylor, "The Spectral Theorem for Self-Adjoint and Unitary Operators," 2018.
- [2] MTTS, "Expository Article on Spectral Theorem," 2023.